

Research Statement

Fuqin Zhou*

July 23, 2026

My research lies at the intersections of supply chain management, food economics, business data science, and emerging technologies, with a focus on how data-driven analytics and intelligent systems can improve the resilience, efficiency, and sustainability of food supply chains. It examines how market dynamics, policy interventions, and supply chain disruptions influence food prices, supply chain resilience, and operational decision-making.

More specifically, my research examines how transportation systems, supply chain disruptions, and emerging technologies shape food prices, market efficiency, and resilience. My dissertation, *Managing Food Supply Chain Affordability and Resilience Through Transportation Analytics*, investigates how transportation factors (i.e., fuel prices, driver availability, and road infrastructure) affect food price dynamics in the U.S., particularly in the context of rising grocery inflation and increasing supply chain uncertainty. Drawing on large-scale datasets, econometric modeling, transportation network analysis, and artificial intelligence, I examine how operational constraints, infrastructure challenges, and external disruptions propagate through food supply chains and how intelligent decision-support systems can enable adaptive coordination and recovery strategies. This research provides a comprehensive framework for understanding disruption impacts and developing more resilient and responsive food supply networks.

In my recently published work in Transportation Research Part E: Logistics and Transportation Review (ABDC: A*), *Food Transportation and Price Impacted by Diesel Price and Truck-driver Shortage pre-, amid and post Pandemic*[1] (co-authored with Dr. Jasmine Chang, Dr. Nesreen El-Rayes, and Dr. Jim Shi, and funded by USDA), we study the U.S. fresh fruit and vegetable market. Specifically, we examine how diesel prices and truck driver shortages influence the market across different pandemic periods. Using panel data on major commodities including apples, potatoes, onions, and tomatoes from 2017 to 2022, we find that rising diesel costs and driver scarcity significantly increase food transportation margins. Among these commodities, potato prices are the most sensitive to changes in transportation

*Martin Tuchman School of Management, New Jersey Institute of Technology. Email: fz24@njit.edu

costs, while tomato prices are the least affected. Overall, fresh produce prices increased between 1.82% and 7.76% in the period after COVID-19. This study identifies key logistical drivers of food prices and provides policy insights for transportation and food security.

In my recently manuscript under revision for International Journal of Production Economics(ABDC: A), *Food on Road: A Novel County-Level Analysis of Road Density and its Impact on Food Prices in the U.S. Market* (co-authored with Dr. Jasmine Chang and Dr. Jim Shi), I serve as the leading author, contributing to research idea formulation, data collection, methodology development, and initial drafting. Building on the previous study of variable transportation costs on labor and fuel, this second paper examines how fixed infrastructure influences regional food prices through its impact on food access and mobility. Although prior studies have examined determinants of food prices, the role of local and regional road networks remains underexplored. To address this gap, the empirical challenge is to isolate the effect of road density from other geographic, demographic, and temporal factors influencing pricing. Consequently, we construct a county-level road infrastructure measure using comprehensive U.S. roads data and compile a unique dataset spanning 2010 to 2020. Our analysis shows that higher road density significantly reduces the cost per meal, with a one mile/mile² increase lowering meal costs by 1.3%, ceteris paribus. Further, mechanism analysis indicates that this effect weakens as store availability increases and even turns positive in densely populated areas. In addition, using spatial econometric models, we identify geographic spillover effects of road density on food prices. Overall, these findings demonstrate how transportation infrastructure influences food mobility and price equity, providing evidence-based insights for planners and policymakers. By creating a novel county-level road density dataset, the study offers spatially and temporally grounded contributions to understanding local food markets.

In a new research stream, I am investigating how autonomous AI agents can enhance coordination and resilience in decentralized food supply chains. In the project I am leading, *AI Agents for Resilient Local Food Systems*(with Dr. Wenbo Wang, Qin Liu, Dr. Jasmine Chang and Dr. Jim Shi)), I propose a multi-agent coordination framework that enables food hubs and their ecosystem partners to collaboratively respond to supply chain disruptions under distributed information. Unlike conventional optimization approaches that assume centralized control, our framework models farmers, buyers, and food hubs as autonomous agents that coordinate through a modular decision architecture. Following disruption detec-

tion, specialized modules perform operational impact assessment, order reallocation, logistics replanning, and recovery negotiation, allowing the hub to dynamically adapt its operations in real time. To facilitate coordination among stakeholders with heterogeneous objectives, we design a privacy-preserving negotiation and concession mechanism that enables agents to reach mutually acceptable recovery decisions while balancing service reliability, food waste reduction, and fairness. In our experiments, we employ large language models to generate realistic disruption scenarios and evaluate how negotiation-driven coordination improves system performance under uncertainty. This project extends my research from understanding the causes and impacts of supply chain disruptions to designing intelligent, AI-enabled systems that proactively support resilient and adaptive decision-making in food supply networks.

In addition to my core research on food prices, transportation, and AI-enabled supply chain resilience, I have also pursued complementary research directions that leverage emerging technologies and advanced analytics to improve food systems. For example, in *Tariff Shocks and Supply Chain Spillovers in the U.S. Tomato Market* (with Dr. Jasmine Chang and Dr. Jim Shi), I examine how trade policy shocks propagate through transportation networks and affect food prices. In another ongoing project, *Supply Chain Graph Neural Networks for Food Price Forecasting* (with Dr. Jasmine Chang and Dr. Jiaying Zhang), we develop a graph neural network framework that captures supply chain dependencies to improve food price forecasting. I have also explored AI-enabled retail applications through the project *AI-Powered XR for Shelf-Life Estimation and Pricing Optimization* (with Dr. Jasmine Chang), which investigates how artificial intelligence and extended reality technologies can support real-time shelf-life estimation and dynamic pricing.

Beyond food systems research, I have also explored the FinTech domain as part of my broader curiosity to apply innovative methods across disciplines. For example, in my paper published in Finance Research Letters (ABDC: A), *How the Economic Policy Uncertainty (EPU) impacts FinTech: The implication of P2P lending markets* [2] (co-authored with Dr. Jasmine Chang and Dr. Jim Shi), we examine how policy uncertainty shapes P2P lending platforms such as LendingClub and Prosper. This project reflects my capability to analyze complex multi-source datasets, apply machine learning techniques, and integrate explainable AI methods, demonstrating my ability to embrace new technologies and adapt them to emerging research contexts.

I have gained substantial experience in securing and managing research funding through projects supported by USDA and the NSF I-Corps program. For the USDA-funded project, I led data collection and analysis, presented findings at conferences, and contributed to peer-reviewed publications, gaining first-hand experience in translating grant support into impactful research outcomes. As the Entrepreneurial Lead in the NSF I-Corps program, I prepared and submitted proposals, completed entrepreneurial training, and conducted over 100 customer interviews, building a well-established network across academia and industry. These experiences have shaped my future research funding plan. I aim to leverage USDA grants to advance research in food systems and supply chain analytics, while pursuing NSF I-Corps and NSF SBIR/STTR programs to further develop and potentially commercialize AI-driven solutions in food.

Looking ahead, my research agenda aims to expand on these themes, continuing to bridge gaps between food supply chain management, food economics, and business data science. By investigating how transportation, food market dynamics, and economic policy interact to influence food networks, I seek to contribute to a more comprehensive understanding of how these forces shape the food business and inflation. For my in-progress work, I am targeting top-tier journals at the intersection of supply chain, operations, and AI applications. They are suitable for journals such as Manufacturing & Service Operations Management (MSOM), Production and Operations Management (POM) and Decision Science Journal(DSJ). Looking forward, I plan to pursue NSF and USDA grants to support these projects, including commercialization efforts for the AI-XR food system through NSF I-Corps and SBIR/STTR programs. These efforts, together with recognition such as the MTSM Award for Excellence in Research, underscore my commitment to advancing innovative, data-driven, and impactful research that bridges theory and practice.

References

- [1] Aichih Jasmine Chang, Fuqin Zhou, Nesreen El-Rayes, and Jim Shi. Food transportation and price impacted by diesel price and truck-driver shortage pre-, amid and post pandemic. *Transportation Research Part E: Logistics and Transportation Review*, 192:103794, 2024.
- [2] Fuqin Zhou, Aichih Jasmine Chang, and Jim Shi. How the economic policy uncertainty (epu) impacts fintech: The implication of p2p lending markets. *Finance Research Letters*, 70:106268, 2024.